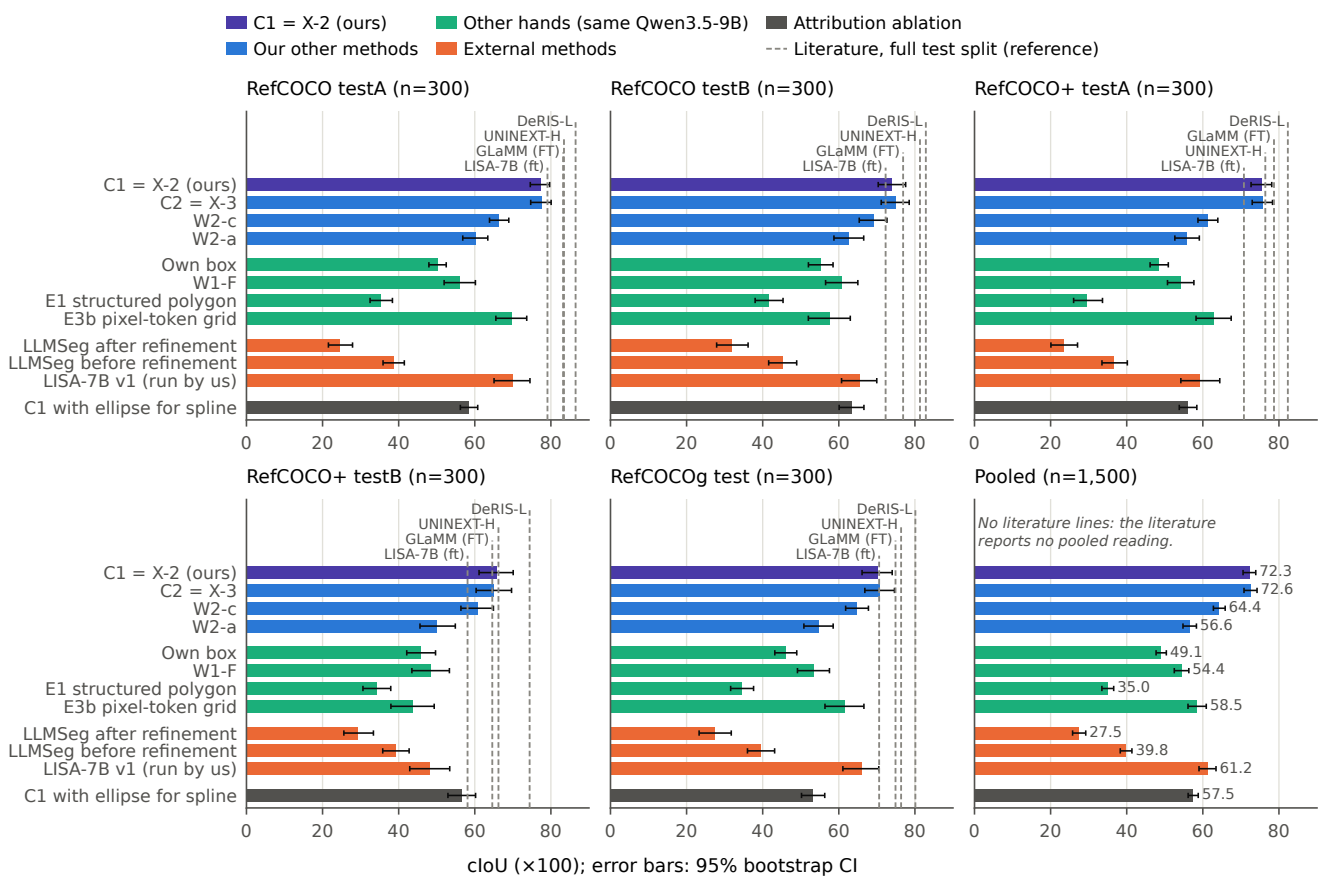
C1 = X-2 (ours)
Other hands (same Qwen3.5-9B)
Attribution ablation
Our other methods
External methods
Literature, full test split (reference)
RefCOCO testA (n=300)
RefCOCO testB (n=300)
RefCOCO+ testA (n=300)
RefCOCO+ testB (n=300)
RefCOCOg test (n=300)
Pooled (n=1,500)
DeRIS-L
UNINEXT-H
GLaMM (FT)
LISA-7B (ft)
C1 = X-2 (ours)
C2 = X-3
W2-c
W2-a
Own box
W1-F
E1 structured polygon
E3b pixel-token grid
LLMSeg after refinement
LLMSeg before refinement
LISA-7B v1 (run by us)
C1 with ellipse for spline
0
20
40
60
80
No literature lines: the literature reports no pooled reading.
72.3
72.6
64.4
56.6
49.1
54.4
35.0
58.5
27.5
39.8
61.2
57.5
cIoU (×100); error bars: 95% bootstrap CI